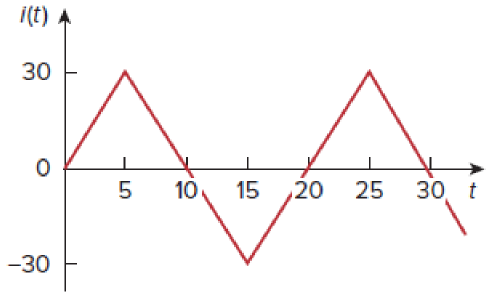
Chapter 11, Problem 29P

Problem

Calculate the effective value of the current waveform in Fig. 11.60 and the average power delivered to a 12-Ω resistor when the current runs through the resistor.

Figure 11.60:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.60 in the textbook.

Determine the expression for $i(t)$.

For $5 < t < 15$,

$$i - 30 = \frac{-30-30}{15-5}(t-5)$$

$$i = 30 - \frac{60}{10}(t-5)$$

$$= 60 - 6t$$

For $15 < t < 25$,

$$i - (-30) = \frac{30-(-30)}{25-15}(t-15)$$

$$i = -30 + \frac{60}{10}(t-15)$$

$$= -120 + 6t$$

Therefore, the current $i(t)$ is expressed as,

$$i(t) = \begin{cases} 60 - 6t, & 5 < t < 15 \\ -120 + 6t, & 15 < t < 25 \end{cases}$$

Step 2 of 3

Calculate the effective value of current.

$$\mathbf{I}_{\text{eff}} = \sqrt{\left(\frac{1}{T}\int_0^T i^2(t)\,dt\right)}$$

Substitute 20 for T in equation.

$$\mathbf{I}_{\text{eff}}^2 = \frac{1}{20}\left(\int_5^{15} i^2(t)\,dt + \int_{15}^{25} i^2(t)\,dt\right)$$

Substitute $(60-6t)$ for $5<t<15$, $(-120+6t)$ for $15<t<25$ in equation.

$$\begin{aligned}
\mathbf{I}_{\text{eff}}^2 &= \frac{1}{20}\left(\int_5^{15}(60-6t)^2\,dt + \int_{15}^{25}(-120+6t)^2\,dt\right)\\
&= \frac{1}{20}\left(\int_5^{15}\left[3600+36t^2-720t\right]dt + \int_{15}^{25}\left[36t^2+14400-1440t\right]dt\right)\\
&= \frac{1}{20}\left(\left[3600t\Big|_5^{15} + 36\frac{t^3}{3}\Big|_5^{15} - 720\frac{t^2}{2}\Big|_5^{15}\right] + \left[6^2\frac{t^3}{3}\Big|_{15}^{25} + 14400t\Big|_{15}^{25} - 1440\frac{t^2}{2}\Big|_{15}^{25}\right]\right)\\
&= \frac{1}{20}\left[60^2[15-5]+12\left[15^3-5^3\right]-360[225-25]+(120)^2[25-15]+12\left[25^3-15^3\right]-720[625-225]\right]
\end{aligned}$$

$$\begin{aligned}
\mathbf{I}_{\text{eff}}^2 &= \frac{1}{20}[36000+39000-72000+144000+147000-288000]\\
&= 300\text{ A}\\
\mathbf{I}_{\text{eff}} &= 17.32\text{ A}
\end{aligned}$$

Therefore, the effective value of current is $\boxed{17.32\text{ A}}$.

Step 3 of 3

Determine the average power delivered through $12\ \Omega$ resistor.

$$P = \mathbf{I}_{\text{eff}}^2 R$$

Substitute 17.32 A for $\mathbf{I}_{\text{eff}}$, $12\ \Omega$ for R in equation.

$$\begin{aligned}
P &= (17.32\text{ A})^2(12\ \Omega)\\
&= (17.32)^2(12)\\
&= 3.599\text{ kW}
\end{aligned}$$

Therefore, the average power delivered through $12\ \Omega$ resistor is $\boxed{3.599\text{ kW}}$.